SUMMER INTERNSHIP REPORT

ON

JAVA PROGRAMMING

AT

CODEAPLHA

SUBMITTED TO:
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
IILM UNIVERSITY, GREATER NOIDA, U.P.

In partial fulfilment of the requirement for the award of the degree of
Bachelor of Technology

SUBMITTED BY:
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Rollno. : 25SCS1003003611
B.Tech – Computer Science & Engineering 3rd Sem

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MINISTRY OF CORPORATE AFFAIRS
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CERTIFICATE
OF COMPLETION

This Certificate Is Proudly Presented To

This Certificate recognizes the successful completion of the CodeAlpha Virtual Internship Program in Java Programming, demonstrating dedication and sincerity throughout the program.
We appreciate the valuable efforts and wish continued success in all future endeavors.

20th July 2026 to 20th August 2026

Alok Kumar Malviya

SwatiSri
CEO & Co-Founder

21st August 2026
Date of Issue

SCAN TO VERIFY THIS CERTIFICATE
Student ID: CA/DF1/207714

CANDIDATE’S DECLARATION

I, **Alok Kumar Malviya,** hereby declare that the internship report titled “Java Programming Internship at CodeAlpha” has been prepared by me to fulfil the academic requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering at IILM University. The report presents the work and learning undertaken during the CodeAlpha Virtual Internship Program in Java Programming from 20 July 2026 to 20 August 2026.

I further declare that the projects and technical work described in this report are based on the work completed during the internship. The information used from external sources has been acknowledged appropriately. I understand that I shall be responsible for any copyright or academic-integrity issues arising from material submitted in this report.

Student Name: Alok Kumar Malviya
Roll No.: 25SCS1003003611

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to CodeAlpha for providing me with the opportunity to complete the Virtual Internship Program in Java Programming. The internship gave me practical exposure to Java programming and helped me apply classroom concepts to project-based tasks.

I am thankful to the CodeAlpha team for providing an environment in which I could strengthen my understanding of Java fundamentals, object-oriented programming, collections, problem solving, and application development. Working on practical projects helped me understand how programming concepts are combined to build functional software applications.

I would also like to thank the faculty members of the School of Computer Science and Engineering, IILM University, Greater Noida, for providing this internship opportunity as part of the academic curriculum.

Finally, I express my gratitude to my family and everyone who supported and encouraged me throughout the internship.

1.1 INTRODUCTION

A summer internship provides students with an opportunity to connect academic learning with practical professional experience. As a Computer Science and Engineering student, the Java Programming internship at CodeAlpha provided practical exposure to programming concepts and project development.

The internship was completed through the CodeAlpha Virtual Internship Program in Java Programming from 20 July 2026 to 20 August 2026. During this period, the focus was on applying Java programming concepts to practical applications and strengthening programming and problem-solving skills.

The project work included a Student Grade Tracker and a Stock Trading Platform. These projects provided experience in input handling, arrays, loops, calculations, classes and objects, encapsulation, ArrayList, transaction processing, portfolio management, and file handling.

1.2 ORGANIZATIONAL PROFILE – CODEALPHA

CodeAlpha is the organization through which the virtual internship program in Java Programming was completed. The internship certificate confirms successful completion of the CodeAlpha Virtual Internship Program in Java Programming and recognizes the participant’s dedication and sincerity during the program.

The internship was conducted virtually and was project-oriented. The Java programming work provided an opportunity to strengthen core programming knowledge and apply object-oriented concepts while developing functional console-based applications.

1.3 PROBLEM STATEMENT

Learning programming only through theoretical concepts can make it difficult for students to understand how individual programming features are combined into complete applications. A practical internship therefore requires the development of applications that use programming logic, data structures, object-oriented design, validation, calculations, and data persistence.

The internship addressed this learning requirement through practical Java projects. The Student Grade Tracker focuses on collecting and processing student marks, while the Stock Trading Platform models a simplified trading environment involving stocks, users, portfolios, transactions, profit/loss, and file-based storage.

1.4 OBJECTIVES OF THE INTERNSHIP

- To strengthen fundamental Java programming skills.
- To apply object-oriented programming concepts in practical applications.
- To improve logical thinking and problem-solving ability.
- To understand the use of arrays and collections for storing data.
- To implement input validation and application-level decision making.
- To understand basic file handling and persistent data storage.
- To gain experience in developing functional console-based Java applications. To improve debugging, testing, and code-organization skills.

1.5 SCOPE OF THE INTERNSHIP

The scope of the internship was focused on Java programming and practical application development. The work involved designing and implementing small-scale applications that demonstrate core programming and object-oriented concepts.

Development of a Student Grade Tracker. Development of a Stock Trading Platform.

- Use of Java classes, objects, methods, constructors, and encapsulation.
- Use of arrays and ArrayList collections.
- Implementation of calculations such as average, highest/lowest marks, investment value and profit/loss. Implementation of transaction records and portfolio operations.
- File handling for saving portfolio information.
- Testing application flow through console-based interaction.

1.6 TECHNOLOGIES AND TOOLS USED

The internship projects were implemented using Java. The technologies and programming concepts used in the supplied project work are summarized below.

- Java Programming Language – primary development language.
- Scanner – used for reading user input in console applications.
- Arrays – used for storing student names and marks.
- ArrayList – used for dynamic collections such as stocks, portfolio items and transactions. Object-Oriented Programming – classes, objects, constructors, methods and encapsulation.
- File Handling – FileWriter and IOException for saving portfolio data.
- Conditional Statements and Loops – used for validation, searching, calculations and menu control. Java Utility Classes – including Date for transaction timestamps.

1.7 PROJECT OVERVIEW AND SYSTEM ARCHITECTURE

Two practical Java applications were developed as part of the project work: Student Grade Tracker and Stock Trading Platform. Both applications use console-based interaction and demonstrate how Java programming constructs can be combined to solve practical problems.

Project 1: Student Grade Tracker

The Student Grade Tracker is a Java application that collects the number of students, their names, and their marks. It stores names and marks in arrays and processes the marks to calculate the total, average, highest marks, and lowest marks. The application then displays a structured student grade report.

The program uses Scanner for input, arrays for data storage, loops for processing, and conditional statements for identifying the highest and lowest marks. The implementation demonstrates basic data processing and console output.

Project 2: Stock Trading Platform

The Stock Trading Platform is a larger Java application designed to simulate basic stock trading operations. The application provides market data and allows a user to buy and sell stocks, view a portfolio, review transaction history, and save portfolio information to a file.

The application is organized using several classes. The Stock class stores stock symbol, company name, and price. PortfolioItem stores the stock, quantity and average buy price. Transaction stores transaction type, stock symbol, quantity, price and date. The User class manages balance, portfolio, transactions, buying, selling, portfolio display and data saving.

The main program creates market stocks, creates a user with an initial balance, and presents a menu for different operations. The implementation uses encapsulation through private fields and getter/setter methods, along with ArrayList for dynamic collections.

SYSTEM ARCHITECTURE

User Input Layer – accepts menu choices, student information, stock symbols and quantities.

Application Logic Layer – performs calculations, validation, buying/selling operations and report generation.

Data Model Layer – uses Java classes such as Stock, PortfolioItem, Transaction and User.

Collection Layer – uses arrays and ArrayList to maintain application data.

Persistence Layer – saves stock portfolio and transaction information to portfolio.txt.

The architecture follows a simple flow: User Input → Validation → Processing → Data Update → Output / File Storage. This structure makes the applications easier to understand and demonstrates separation of responsibilities among classes and methods.

1.8 METHODOLOGY

The methodology followed during the internship was practical and project-oriented. Each application was developed by first understanding the required features, identifying the data to be stored, selecting suitable Java constructs, implementing the functionality, and testing the resulting program.

- Step 1: Requirement Understanding – identify the purpose and features of each application. Step 2: Program Design – identify required data, classes, methods, and processing logic.
- Step 3: Implementation – write Java code using appropriate programming constructs.
- Step 4: Validation – check invalid inputs such as non-positive quantities and unavailable stocks. Step 5: Testing – execute different menu options and verify calculated outputs.

- Step 6: Data Management – maintain collections and, where required, save information to a file.
- Step 7: Review – examine the program structure and improve understanding of Java and OOP concepts.

1.9 WORK PERFORMED AND RESPONSIBILITIES

The internship work involved practical implementation and understanding of Java programming concepts through project development.

- Designed and implemented Java console applications.
- Worked with Scanner for user input.
- Used arrays for student data processing.
- Implemented calculations for average, highest and lowest marks. Designed classes for stock, portfolio item, transaction and user entities. Implemented buy and sell stock functionality.
- Implemented portfolio valuation and profit/loss calculation. Maintained transaction history.
- Implemented file-based portfolio saving.
- Applied validation and conditional logic to handle invalid operations. Practised organizing functionality into methods and classes.
- Tested application workflows through multiple console interactions.

1.10 EXPECTED OUTCOMES

The internship was expected to provide practical understanding of Java programming beyond classroom exercises. By completing project-based tasks, the internship aimed to improve programming logic, object-oriented design, data handling, and problem-solving skills.

- Improved understanding of Java fundamentals.
- Better application of OOP concepts.
- Ability to work with arrays and collections.
- Improved understanding of file handling.
- Experience in developing menu-driven applications.
- Improved debugging and logical reasoning.
- Greater confidence in converting requirements into working

1.11 LEARNING OUTCOMES

The internship helped strengthen both technical and professional learning. The Student Grade Tracker provided practice with basic data structures, loops, calculations and input handling. The Stock Trading Platform provided deeper exposure to object-oriented programming, collections, validation, transaction processing and file handling.

Java programming and syntax Object-oriented programming Classes and objects Encapsulation

- Constructors and methods
- Arrays and ArrayList
- Input validation
- File handling and exception handling Problem solving and logical reasoning Program organization and testing

1.12 CONCLUSION

The CodeAlpha Virtual Internship Program in Java Programming provided valuable practical exposure to application development. The internship allowed academic Java concepts to be applied in functional projects, making the learning process more practical and structured.

The Student Grade Tracker strengthened understanding of arrays, loops, input handling and basic data processing. The Stock Trading Platform provided broader experience with object-oriented design, collections, portfolio operations, transactions, calculations and file storage.

Overall, the internship contributed to the development of programming, problem-solving, application-design and technical skills. It also provided a foundation for undertaking more advanced software development projects in the future.

BIBLIOGRAPHY / REFERENCES

- CodeAlpha Virtual Internship Program in Java Programming – Internship Certificate, dated 21 August 2026. Java programming concepts and standard Java library documentation used during project implementation. Project source code: Student Grade Tracker.
- Project source code: Stock Trading Platform.